

Elias Hossain

Trustworthy & Safe Large Language Models · AI Agents · Alignment & Uncertainty

Orlando, FL | mdelias.hossain@ucf.edu | [Google Scholar](#) | [Website](#) | [GitHub](#) | [LinkedIn](#)

Available for Summer 2027 internships / co-op (PhD coursework complete) | F-1 international student, CPT-eligible.

RESEARCH SUMMARY

PhD researcher building **trustworthy and safe large language models and AI agents**: LLM alignment under corrupted preference supervision (RLHF/DPO), calibrated uncertainty for foundation models, runtime safety for tool-using agents, and inference-time control of frozen policies. First author on a paper accepted to the **ICLR 2026 Trustworthy AI Workshop**, with further first-author work under review at **NeurIPS** and **ACL**.

943 Google Scholar citations | **h-index 12** | **14+ open-source repositories** | **4 public benchmarks**

EDUCATION

Ph.D., Industrial Engineering (CS concentration), University of Central Florida Aug 2025 – Present
Advisor: Dr. Niloofar Yousefi. Passed PhD qualifying exam. Expected graduation Spring /Fall 2028.

M.S., Computer Science (Research), Mississippi State University Jan 2024 – Jul 2025
Advisors: Dr. Andy D. Perkins, Dr. Shahram Rahimi.

B.S., Software Engineering, Daffodil International University 2016 – 2020

EXPERIENCE

Graduate Research Assistant, UCF Complex Adaptive Systems Lab Aug 2025 – Present

- Lead an independent research program on trustworthy LLMs and agent safety, advised by **Dr. Niloofar Yousefi** and collaborating with **Prof. Ser-Nam Lim**, former Senior Research Scientist Manager at Meta AI; first-author work accepted at ICLR 2026 (Trustworthy AI Workshop) and under review at NeurIPS and ACL.
- Designed and implemented **NEXUS**, a runtime safety monitor for tool-using LLM agents, raising intervention accuracy +**27.3 points** over rule-only baselines; released the code and **4 public benchmarks (74,590+ examples)**.

Graduate Research Assistant, Mississippi State University May 2025 – Jul 2025

- Built and deployed **BarkPlug V.2**, a production RAG chatbot (FAISS retrieval; evaluated with RAGAS and the System Usability Scale); also developed a contrastive-learning framework for anomaly detection over large-scale cybersecurity event logs.

Software Engineer, REVE Systems, Dhaka, Bangladesh Dec 2022 – Nov 2023

- Trained and deployed production speech-recognition models (DeepSpeech, Wav2Vec 2.0): **20% lower Word Error Rate, 15% faster inference**; shipped a voice extension for national e-governance services.

Senior AI/ML Researcher (Remote), Time Research & Innovation, Portsmouth, UK Oct 2020 – Nov 2022

- Designed and trained deep-learning architectures for COVID-19 diagnosis from X-ray/CT imagery; co-built a UK telemedicine platform.

SELECTED PUBLICATIONS

Published or accepted; first author unless noted. Full list on Google Scholar.

- Fault-Tolerant Preference Alignment via Multi-Agent Verification. **Elias Hossain** et al. *ICLR 2026 Trustworthy AI Workshop*. [\[link\]](#)
- Safe and Scalable Collaboration in Multiagent LLM Systems: A Comprehensive Review. **Elias Hossain** et al. *IEEE Trans. on Systems, Man, and Cybernetics: Systems*, 2026. [\[link\]](#)
- BIOGEN: Evidence-Grounded Multi-Agent Reasoning for Transcriptomic Interpretation in Antimicrobial Resistance. **Elias Hossain**, M. Shoeibi, I. Garibay, N. Yousefi. *Frontiers in Bioinformatics*, 2026. [\[link\]](#)
- R-GAT: Cancer Document Classification Leveraging a Graph-Based Residual Network for Scenarios with Limited Data. **Elias Hossain**, T. Nuzhat, S. Masum, S. Rahimi, N. Amiri Golilarz. *Scientific Reports* (Nature Portfolio), 2026. [\[link\]](#)
- Natural Language Processing in Electronic Health Records in Relation to Healthcare Decision-Making: A Systematic Review. **Elias Hossain**, R. Rana, N. Higgins, J. Soar, P. D. Barua, A. R. Pisani, K. Turner. *Computers in Biology and Medicine*, 2023. **(513 citations)** [\[link\]](#)

MANUSCRIPTS UNDER REVIEW

First author unless noted; venues withheld while under review.

- NEXUS: Structured Runtime Safety for Tool-Using LLM Agents. **Elias Hossain** et al. *Under review*, 2026. [Code]
- When Policies Cannot Be Retrained: A Unified Closed-Form View of Post-Training Steering in Offline RL. **Elias Hossain**, M. J. I. Basher, I. Garibay, O. Garibay, N. Yousefi. *Under review*, 2026. [arXiv]
- UAT-LITE: Inference-Time Uncertainty-Aware Attention for Pretrained Transformers. **Elias Hossain**, S. Roy Dipta, S. Neupane, R. Rana, R. Shwartz-Ziv, I. Garibay, N. Yousefi. *Under review*, 2026. [arXiv]
- CAREBENCH: When Correct Diagnoses Come from Unsafe Reasoning. **Elias Hossain**, T. Nuzhat Ornee, S. Roy Dipta, I. Garibay, A. Rivera-Sepulveda, N. Yousefi. *Under review*, 2026.
- Right Knowledge, Wrong Answer: Characterizing Parametric Temporal Conflict in Open-Weight Language Models. **Elias Hossain**, S. Saha, U. C. Biswas, S. S. Jennifer. *Under review*, 2026. [arXiv]
- BioDivergence: A Benchmark and Evaluation Framework for Hidden Contextual Contradictions in Biomedical Abstracts. **Elias Hossain**, S. S. Jennifer, S. A. Bushra, N. Yousefi. *Under review*, 2026. [arXiv]
- MedBayes-Lite: A Clinical Uncertainty Governance Layer for Risk-Aware Medical Decision Support. **Elias Hossain**, M. M. H. Nipu, M. Sheikh, T. Nuzhat, R. Rana, S. Neupane, B. W. Schuller, N. Yousefi. *Under review*, 2026. [arXiv]
- MotifRole-Diff: Risk-Optimal Role-Aware Corruption for Masked Molecular Graph Diffusion. T. Nuzhat Ornee, **Elias Hossain**, I. Garibay, N. Yousefi. *Under review*, 2026.

SELECTED PROJECTS, BENCHMARKS & OPEN SOURCE

Reproducible code, datasets, and evaluation pipelines released for all major projects. huggingface.co/EliasHossain | github.com/eliashossain001

- **CorrFilter** (with **Prof. Ser-Nam Lim**, former Senior Research Scientist Manager at Meta AI): showed that LLM-judge banks used for alignment-data filtering fail dependently, so consensus filtering degrades as judges correlate (effective independent judges near 5 of 16); built a calibration-based adaptive filter that reduces false retention by up to **6 points**.
- **Released 4 public benchmarks (74,590+ examples)** on Hugging Face: schema-constrained biomedical extraction ([nanobubbleeval](#), 51.6K), biomedical contradiction NLI ([BioDivergence](#), 11.9K), knowledge-conflict temporal QA ([ptc-benchmark](#), 9.25K), and cancer-abstract classification ([CancerAbstracts](#)).
- **Inference-time steering of frozen policies**: [PoE-Composition](#) (product-of-experts refinement for offline RL, evaluated across D4RL) and [Temporal Attractor Steering](#) (single-layer intervention corrects **72–85%** of temporal-conflict answers at **85–99%** retention on non-conflicting queries across Qwen/Mistral/Llama, no retraining).
- **LLM systems from the ground up**: [Qwen-style 0.6B transformer built from scratch in PyTorch](#) (tokenizer, attention, training loop), plus [LoRA/QLoRA \(PEFT\) multi-GPU fine-tuning](#) and [continual pretraining](#) for domain adaptation without catastrophic forgetting.

TECHNICAL SKILLS

- **LLMs & Alignment**: fine-tuning & alignment (RLHF, DPO, SFT); PEFT (LoRA/QLoRA); AI agents & tool use; inference-time steering/control of frozen models; continual pretraining; offline RL; multimodal modeling.
- **Trustworthy AI & Evaluation**: uncertainty quantification, calibration, Bayesian learning; LLM evaluation & benchmarking; runtime safety; adversarial robustness (prompt injection, data poisoning); graph neural networks.
- **Systems & Tools**: Python, R, SQL, Bash; PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn; multi-GPU / distributed training; FAISS, Chroma, RAG; Apache Spark; Docker, Git, Azure, GCP.

RECOGNITION & SERVICE

- **Talks**: ICLR 2026 Trustworthy AI Workshop; MoML, MIT Jameel Clinic (2025); Amazon Trusted AI Symposium poster (2026).
- **Awards**: UCF Graduate Presentation Fellowship (2025); Finalist, Dr. Pradeep P. Thevannoor Innovation Awards (2019).
- **Service & Teaching**: Reviewer for ICLR Trustworthy AI Workshop (2026) and 7 journals (Nature Portfolio, Elsevier, PLOS, SAGE, Springer); Graduate TA in ML/AI.